# AI-BASED CLASSIFICATION OF INDICATORS FOR VISUAL ADAPTATION OF INSTITUTIONAL DASHBOARDS

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## Objectives

Institutional *dashboards* are widely adopted resources to support academic management through the visualization of indicators such as dropout and performance. However, the simultaneous presentation of multiple data can generate visual overload [4], such that inadequate visual organization hinders the identification of priority metrics, compromising support for decision-making in academic management. Although well-established principles of visual hierarchy exist [3], there are still few proposals that integrate an AI model into the automatic classification of indicator criticality, in order to adapt its presentation in institutional dashboards. This raises the question: In what way can AI-based automated classification of indicator criticality guide the application of visual hierarchy in institutional dashboards? This work aims to develop a prototype dashboard with automated criticality classification and to carry out a preliminary technical analysis of its visual hierarchy, proposing as a contribution a *pipeline* that automatically transforms these classifications into metadata consumed by the frontend to adapt the visual hierarchy.

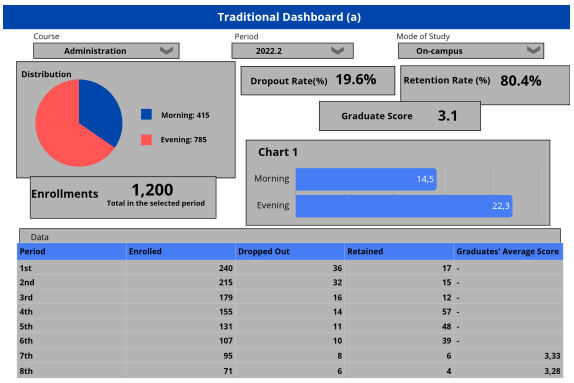
## Methods and Procedures

The development was based on an automated *pipeline* in which 20 simulated academic *KPIs* (including name, value, and reference average) were pre-processed into *JSON* and sent to the Gemini 1.5 *Flash* [2] (temperature 0.1). The structured *prompt* classified the indicators as normal, attention, or critical, generating visual metadata. The criticality rule defined “Alert” for deviation above 10% from the institutional average, “Attention” between 5% and 10%, and “Normal” below 5%, validated by the percentage agreement between the AI and a reference answer key of the 20 *KPIs*. The *frontend* consumes the metadata to dynamically reorganize the elements into a sequence composed of overview, critical alerts, and details, guiding the reading of context toward urgencies. Data integrity was maintained via *hashing* of identifying keys, simulating compliance with the LGPD [1]. Finally, a technical analysis ensured that the states were represented by color, label, and icon, meeting visual accessibility guidelines [5].

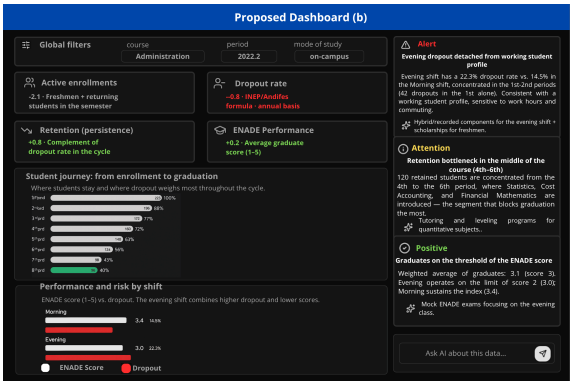
## Results

The technical analysis compared the conventional homogeneous interface (Figure 1a) to the AI-guided adaptive model (Figure 1b). Of the 20 *KPIs* analyzed, six had previously been classified as critical by the researchers for exceeding the simulated institutional limits. The AI model showed 100% agreement with this reference, generating metadata that dynamically reconfigured 8 *cards*, 2 charts, and 8 *badges.* In the proposed version, the critical deviation for nighttime dropout (22.3%) was moved to the top of the panel, receiving greater visual contrast through

the simultaneous application of alert color, icon, and *badge* text. The criticality states were represented in a multimodal way (color, text, and icon) to meet accessibility criteria, demonstrating the technical feasibility of dynamically injecting styles into the interface based on the AI model's inferences. Quantitative measurement of contrast ratios is planned for future validation stages.




(a)




(b)

Figure 1: (a) Traditional Dashboard with charts only; (b) Proposed Dashboard with AI integration (authors, 2026).

## Conclusions

The integration of a language model for automated criticality classification proves to be technically feasible for organizing the visual hierarchy in institutional dashboards. The results demonstrate that the *pipeline* developed is capable of converting indicator analyses into style metadata without compromising the original statistical integrity, allowing urgent information to be highlighted autonomously. This preliminary analysis suggests that multimodal visual elements (color, icon, and text) have the potential to organize information density and reduce visual overload, although restricted to the experimental environment investigated. As the study focused on the engineering of the solution, validation of decision-making effectiveness through user testing and quantitative measurement of contrast ratios remain necessary future steps to confirm the intended perception and usability gains in academic management. The authors declare no conflict of interest. Author Lara Fabyã dos Santos worked on the development of the *frontend* adaptive frontend and on the collection of experimental computational metrics, and author Edsamuel da Silva Rocha conceived the study and developed the *pipeline* and AI engineering. All authors reviewed and approved the final version of the manuscript.

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